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Original Article

Sensitivity analysis for clustered observational studies with an application to the effectiveness of magnet nursing hospitals

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Abstract

In a clustered observational study, treatment is assigned to groups and all units within the group are exposed to the treatment. Here, we use a clustered observational study (COS) design to estimate the effectiveness of Magnet Nursing certificates for emergency surgery patients. Recent research has introduced specialized weighting estimators for the COS design that balance baseline covariates at the unit and cluster level. These methods allow researchers to adjust for observed confounders but are sensitive to unobserved confounding. In this paper, we develop new sensitivity analysis methods tailored to weighting estimators for COS designs. We provide several key contributions. First, we introduce a key bias decomposition, tailored to the specific confounding structure that arises in a COS. Second, we develop a sensitivity framework for weighted COS designs that constrain the error in the underlying weights. We introduce both a marginal sensitivity model and a variance-based sensitivity model, and extend both to accommodate multiple estimands. Finally, we propose amplification and benchmarking methods to better interpret the results. Throughout, we illustrate our proposed methods by analysing the effectiveness of Magnet nursing hospitals.

Keywords: clustered data, clustered observational study, sensitivity analysis

1 Introduction

1.1 Magnet nursing hospitals

One key part of health services research is focused on how health care delivery systems affect patient outcomes. For example, do hospitals with superior nursing care have better patient outcomes? One strand of this research has considered whether hospitals with a Magnet Nursing certification have better outcomes for patients. The American Nurses Credentialing Center (ANCC) certifies that hospitals have excellent nursing environments through the Magnet Recognition Programme. Hospitals complete an arduous application process and then undergo a peer review that takes approximately one year to complete before they are rewarded with a Magnet certification (Aiken et al., 1994). Magnet nursing certification has become a widely recognized mark of superior nursing and has become a criterion for ranking in the US News and World Report's Best Hospitals (Kutney-Lee et al., 2015). Magnet hospitals were first recognized for their ability to attract and keep nurses by fostering a supportive and positive workplace (Hospitals, 1983). These hospitals were empirically distinguished by their decentralized, flat organizational structures, an empowered frontline staff actively participating in decision-making and shared

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governance, a culture fostering interprofessional teamwork, and transformational clinical leadership (McHugh et al., 2013). In terms of whether Magnet hospitals have better patient outcomes, the extant literature has shown that hospitals with Magnet Nursing certification tend to have lower mortality and failure-to-rescue rates as well as better value (lower mortality with similar costs) (Aiken et al., 1994; Friese et al., 2015; Lake et al., 2012; McHugh et al., 2013; Mitchell & Shortell, 1997; Silber et al., 2016)

Past research on Magnet Nursing hospitals has focused on a range of medical conditions, but has yet to study outcomes for emergency general surgery (EGS) patients. Emergency general surgery refers to medical emergencies where the injury is internal—such as a burst appendix or emergency resection for diverticulitis. For the purposes of health service research, researchers have designated 51 medical conditions as EGS conditions that are distinct from trauma conditions (Gale et al., 2014; Shafi et al., 2013). Care for patients with EGS conditions includes both surgery and a variety of non-operative measures including observation, minimally invasive procedures, medication, and supportive care. EGS conditions are common with an estimated 3–4 million patients per year that account for more than 800,000 operations in the United States alone (Gale et al., 2014; Havens et al., 2015; Ogola et al., 2015; Scott et al., 2016). A growing literature has focused on the comparative effectiveness of operative vs non-operative care for various EGS conditions (Grieve et al., 2023; Hutchings et al., 2022; Kaufman et al., 2022; Moler-Zapata et al., 2022). Here, we study whether hospitals with Magnet certification have better outcomes for patients with EGS conditions.

In our study, we use a dataset study that merges the American Medical Association (AMA) Physician Masterfile with all-payer hospital discharge claims from Florida and Pennsylvania in 2012–2013. The study population includes all patients admitted for emergency or urgent inpatient care for the 51 recorded EGS conditions. The data also include indicators for the 51 different EGS conditions. The data contain a set of demographic measures including indicators for race, sex, age, and insurance type. The data include indicators for severe sepsis or septic shock and pre-existing disability. Next, there are indicators for 31 comorbidities based on Elixhauser indices (Elixhauser et al., 1998). Finally, there is an indicator for whether the patient had surgery or not. The data also contain a more limited set of hospital-level covariates. Hospital-level measures include hospital patient volume, the proportion of emergency admissions, the proportion of patients that had surgery, and the proportion of patients with Medicaid.

Our focus here is on two different outcome measures. The first outcome is an indicator for the presence of an adverse event following treatment for an EGS condition. The measure of adverse events is an indicator that is 1 if a patient either died, had a complication, or had a prolonged length of stay. The second outcome is a measure of failure to rescue (FTR)—measured as death after a complication—and is intended to measure a failure or delay in recognizing and responding to a patient experiencing complications from hospitalization (Silber et al., 1992). Across these two states, we have 38 hospitals with Magnet status and 368 hospitals that we designate as controls. In the Magnet hospitals, there are 469,579 EGS patients, and in the control hospitals there are 1,990,315 EGS patients. In Magnet hospitals, 32% of EGS patients had an adverse event, and in the control hospitals, 27% of patients had an adverse event. For Magnet hospitals, the FTR percentage was 29% and for non-Magnet hospitals the FTR percentage was 27%. These differences are statistically significant but do not include adjustment for baseline confounders and may reflect key differences in the type of hospitals that receive Magnet status rather than any effect of Magnet status itself. Next, we outline a specific research design that we use to answer the question of whether Magnet hospitals offer superior care to EGS patients.

1.2 Clustered treatment assignment

One common research design in education and health services research is the clustered observational study (COS). In a COS design, an intervention is allocated to groups or clusters of units—e.g. to hospitals or schools—while outcomes of interest are measured at the unit level—e.g. patients or students. There are two critical aspects of treatment assignment in a COS design (Page et al., 2020; Pimentel et al., 2018; Ye et al., 2024). The first critical feature of the COS treatment assignment processes is that all or none of the units within a cluster are exposed to the treatment of interest. The second feature is that treatment assignment is non-random. In health services research, COS

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designs arise when interventions are applied to entire hospitals or physicians. Such is the case with Magnet Nursing. Here, the intervention is applied to all patients that receive care from a Magnet Nursing hospital and withheld from all patients that visit a non-Magnet hospital. In addition, hospitals are clearly not allocated to Magnet status via a random assignment given that the certification programme is highly purposeful.

Given non-random assignment of the treatment in a COS, differences in outcomes may reflect pretreatment differences in treated and control groups rather than actual treatment effects (Hansen et al., 2014). For example, Magnet hospitals tend to have larger patient volumes and often are academic medical centres. As such, these factors may account for better patient outcomes rather than any differences in nursing care. As such, consistent with any observational study, we must adjust for observed confounders to make Magnet and non-Magnet hospitals comparable. Statistical adjustment can be done via outcome modelling with regression models. Recent work, however, has focused on the development of specialized forms of statistical adjustment for observed confounders including matching (Keele & Zubizarreta, 2017; Pimentel et al., 2018), the parametric g-formula (Ye et al., 2024), and balancing weights (Ben-Michael et al., 2024).

However, all these methods share the same key assumption: that all confounders are measured. That is, we must assume that there are no unobserved factors that influence both treatment at a Magnet hospital and the outcome. Critically, this assumption is untestable, but it can be probed via a sensitivity analysis. A sensitivity analysis is designed to *quantify* the degree to which a key identifying assumption must be violated for an original conclusion to be reversed (Rosenbaum, 2002, ch. 4). Hansen et al. (2014) first outlined a sensitivity analysis for COS designs, but it has two key limitations. First, it is only designed for use with matching estimators, but recent work has developed weighting estimators that can substantially improve on matching methods for adjustment in COS designs (Ben-Michael et al., 2024). Second, this form of sensitivity analysis uses a set of worst-case bounds that are often overly conservative (Huang & Pimentel, 2025).

1.3 Main contributions and related work

In this article, we develop sensitivity analysis methods for COS designs based on weighting estimators. Currently, a variety of approaches have been developed to conduct sensitivity analyses for weighting estimators (Huang & Pimentel, 2025; Huang et al., 2025; Tan, 2006; Zhao et al., 2019). However, these methods assume that that treatment assignment is at the unit level; as such, they are not directly applicable to COS applications. The methods we develop are tailored to the balancing weight methods introduced in Ben-Michael et al. (2024), which adjust for observed confounders in contexts where treatment is assigned to clusters.

Our contribution is three-fold. First, we derive a bias decomposition framework for clustered treatment assignments. In this framework, we formalize how omitted covariates at either the unit or cluster level can bias treatment effect estimates in the COS design. We highlight the fact that the magnitude of the bias differs depending on whether the omitted covariate is measured at the unit or cluster level. Specifically, we show that omitting cluster-level covariates will tend to amplify bias. Second, we develop a sensitivity framework for weighted COS designs that constrain the error that arises in the underlying weights. Specifically, we adapt the marginal sensitivity model (Tan, 2006; Zhao et al., 2019) and the variance-based sensitivity model (Huang & Pimentel, 2025; Huang et al., 2025) to account for the COS design. While these approaches have been introduced for the unit level treatment assignment setting, the extension for clustered treatment assignments has not yet been explored. Furthermore, by bounding the error that arises in the underlying weights, the sensitivity models have the flexibility to accommodate multiple estimands beyond the ATE and ATT, such as the average treatment effect in the overlap population, i.e. the ATO (Li et al., 2018). This extension is critical to our analysis of Magnet nursing hospitals, since we find limited overlap between the treated and control covariates at the hospital level. In addition, this represents the first extension of sensitivity analysis methods to the ATO estimand.

Finally, to account for the additional complexity in the COS design of having omitted confounders at both the individual and cluster level, we introduce an amplification approach that allows researchers to decompose a single sensitivity parameter into two components: one that controls for the confounding effect from omitting an individual-level confounder, and one that controls for the confounding effect from omitting a cluster-level confounder. We introduce benchmarking

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approaches that, when used in conjunction with the amplification procedure, can allow researchers to leverage observed confounders to reason about the plausibility of omitted confounders.

Our article proceeds as follows. In Section 2, we provide a formal description of the COS design using potential outcomes notation. We also review extant balancing weight methods. In Section 3, we conduct our initial analysis of Magnet hospitals. Since a sensitivity analysis follows estimation of treatment effects, we outline the initial evidence for that Magnet nursing improves outcomes for EGS patients. In Section 4, we derive a bias decomposition for clustered treatment assignment. In Section 5, we develop two sensitivity analysis models for the COS design and apply them to the Magnet nursing estimates. We then evaluate these methods via simulation in Section 6. In Section 7, we develop tools for interpretation and demonstrate their use with the Magnet nursing sensitivity analysis.

2 The COS design

2.1 Notation and estimands

We consider a setup with m hospitals, with n_ℓ patients in hospital ℓ, and $n = \sum_{\ell=1}^{m} n_\ell$ total patients indexed as $i = 1, \ldots, n$. We denote the treatment status of cluster ℓ as $A_\ell \in \{0, 1\}$, $n_t \equiv \sum_{\ell=1}^{m} A_\ell n_\ell$ as the total number of treated units, and $n_c \equiv n - n_t$ as the total number of control units. Here, when $A_\ell = 1$, hospital ℓ has magnet status, and when $A_\ell = 0$ it does not. Each collection of possible treatment status vectors $\boldsymbol{a} = (a_1, \ldots, a_m) \in \{0, 1\}^m$ is associated with a vector of potential *cluster assignments*, $\boldsymbol{J}(\boldsymbol{a}) = (J_1(\boldsymbol{a}), \ldots, J_n(\boldsymbol{a})) \in \{1, \ldots, m\}^n$, where $J_i(\boldsymbol{a}) \in \{1, \ldots, m\}$ denotes the cluster that unit i would belong to under the overall treatment allocation vector $\boldsymbol{a}$. We denote the observed treatment allocation vector as $\boldsymbol{A} = (A_1, \ldots, A_m)$ and the corresponding observed cluster assignment vector as $\boldsymbol{J} = \boldsymbol{J}(\boldsymbol{A}) = (J_1(\boldsymbol{A}), \ldots, J_n(\boldsymbol{A}))$.

In principle, each unit i has a *potential outcome* $Y_i(\boldsymbol{a})$ that corresponds to the outcome that would be observed if the overall cluster-level treatment assignment vector is $\boldsymbol{a}$. This leads to an unwieldy 2^m possible potential outcomes. To simplify the setup and be able to estimate reasonable forms of treatment effects, we follow Ye et al. (2024) and assume that patient i's potential outcome only depends on the overall vector of hospital assignments $\boldsymbol{J}(\boldsymbol{a})$ induced by treatment assignment vector $\boldsymbol{a}$ and the treatment status of its own hospital $a_{J_i(\boldsymbol{a})}$, i.e. that $Y_i(\boldsymbol{a}) = Y_i(\boldsymbol{a}')$ if $\boldsymbol{J}(\boldsymbol{a}) = \boldsymbol{J}(\boldsymbol{a}')$ and $a_{J_i(\boldsymbol{a})} = a'_{J_i(\boldsymbol{a}')}$. This assumption limits treatment effect spillovers to only operate through the fact that individual hospital assignments can depend on the treatment assignment of all hospitals. With this, we will denote unit i's potential outcome with treatment assignment vector $\boldsymbol{a}$ as $Y_i(a_{J_i(\boldsymbol{a})}, \boldsymbol{J}(\boldsymbol{a}))$. In this setup, the set of potential outcomes $\{Y_i(a, \boldsymbol{j}) \mid a \in \{0, 1\}, \boldsymbol{j} \in \{1, \ldots, m\}^n\}$ that correspond to all possible outcomes for unit i across all combinations of potential treatment and cluster assignment vector are considered to be i.i.d, though the observed outcomes can be statistically dependent within clusters due to the treatment and cluster assignments, as we discuss below.

In this setup, the observed outcome Y_i for unit i is the potential outcome under the observed treatment and cluster assignment vectors, $\boldsymbol{A}$ and $\boldsymbol{J}$, so that $Y_i = Y_i(A_{J_i}, \boldsymbol{J})$. Along with the outcomes, we also observe unit-level covariates $X_1, \ldots, X_n \in \mathcal{X}$ and cluster-level covariates $K_1, \ldots, K_m \in \mathcal{K}$. In our application—and many others—cluster-level covariates include cluster-level aggregations and summaries of the individual-level covariates (e.g. the number of patients seen in the hospital, or the proportion of patients on Medicaid). This means that the cluster-level covariates are affected by cluster assignment, and so cluster ℓ has *potential cluster-level covariates* $K_\ell(\boldsymbol{J}(a))$; the observed cluster-level covariates for cluster ℓ are then the potential value under the observed cluster assignment $K_\ell = K_\ell(\boldsymbol{J}(\boldsymbol{a}))$. Taken together, our observed data consists of tuples $(X_i, J_i, K_{J_i}, A_{J_i}, Y_i)$. We will assume that the observed outcomes are statistically independent across clusters—i.e. that Y_i and $Y_{i'}$ are independent if $J_i \neq J_{i'}$—but may be dependent within clusters.

As additional pieces of notation, we use $\mathcal{R} = \{\mathcal{K}, \mathcal{X}\}$ to represent the combined set of patient and hospital-level covariates. We also define two conditional expectations of the observed outcomes. We denote $m(a, k) \equiv \mathbb{E}[Y_i \mid A_{J_i} = a, K_{J_i} = k]$ as the expected outcome for unit i, conditioned on its cluster having treatment status a and covariates k. We use $m(a, k, x) \equiv \mathbb{E}[Y_i \mid A_{J_i} = a, K_{J_i} = k, X = x]$ to denote the expected outcome if we add more information and additionally condition on unit-level covariates x. Finally, we let $\pi(k) = P(A_\ell = 1 \mid K_\ell = k)$ represent the *cluster-level propensity score* that measures the conditional probability that cluster ℓ is assigned to treatment, given

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the cluster-level covariates, and $\pi(k, x) \equiv \Pr(A_{J_i} = 1 \mid K_{J_i} = k, X_i = x)$ represent the *unit-level propensity score* that measures the conditional probability that an individual i is assigned to a cluster that is assigned to treatment, given the cluster- and individual-level covariates.

Generally, we focus on the average treatment effect for the treated units (ATT) under the observed cluster assignments $\boldsymbol{J}$ as the target causal estimand. We write the ATT in a COS design as[1]

$$\tau^{\text{ATT}} \equiv \underbrace{\mathbb{E}\big[Y_i(1, \boldsymbol{J}) \mid A_{J_i} = 1\big]}_{\mu_1} - \underbrace{\mathbb{E}\big[Y_i(0, \boldsymbol{J}) \mid A_{J_i} = 1\big]}_{\mu_0}. \tag{1}$$

In our COS design, the ATT estimand measures the effect of treatment for patients in Magnet hospitals, keeping the cluster assignments fixed. For units in the treated group, the potential outcome corresponds to the observed outcome, $Y_i(A_{J_i}, \boldsymbol{J}) = Y_i$. Therefore, the first term is equal to the expected value of the observed outcome among the treated group, $\mu_1 = \mathbb{E}[Y_i \mid A_{J_i} = 1]$, and it can be estimated using the sample average outcome for units in the treated clusters: $\hat{\mu}_1 = \frac{1}{n_t}\sum_{i=1}^{n} A_{J_i} Y_i$. See Ye et al. (2024) for a detailed treatment on identification in COS designs. The challenge is in identifying and estimating μ_0, the mean counterfactual outcome if those clusters had in fact been assigned to control. Below, we review identification conditions for μ_0 under the ATT. However, our methodology readily extends to other common estimands such as the average treatment effect (ATE). Specifically, we also ensure that our methods extend to one estimand known as the average treatment effect for the overlap population (ATO) (Li et al., 2018). In the COS setting, we define the ATO as

$$\tau^{\text{ATO}} \equiv \frac{\mathbb{E}\big[\pi(K_{J_i}, X_i)(1 - \pi(K_{J_i}, X_i))\big(Y_i(1, \boldsymbol{J}) - Y_i(0, \boldsymbol{J})\big)\big]}{\mathbb{E}[\pi(K_{J_i}, X_i)(1 - \pi(K_{J_i}, X_i))]}. \tag{2}$$

This estimand measures the treatment effect for the marginal population of units that have close to equal odds of being assigned to a cluster that does or does not get assigned treatment, or units at 'equipoise' (Thomas et al., 2020). The ATO is a helpful estimand when some units are almost deterministically assigned to a cluster that does or does not receive treatment (i.e. $\pi(K_{J_i}, X_i)$ is close to or exactly equal to 0 or 1): it is difficult or impossible to estimate treatment effects for such units, but they are downweighted in the construction of the ATO. Therefore, for the ATO, the required overlap assumption is substantially weaker than for the ATT: it only requires overlap in the unit- and cluster-level covariates for *some* units in the population. In our context, the ATO measures the effect of a Magnet hospital for the population of treated hospitals that could appear in the control group with substantial probability. That is, the ATO weights place greater emphasis on the hospitals and patients with propensity scores close to 1/2 that could be in either group. In turn, the ATO places relatively less weight on hospitals and patients with propensity scores close to 0 or 1 (Li et al., 2018). As such, the ATO may be policy relevant because it measures the effect for those hospitals close to the margin of Magnet status and gives us an indication of what might happen if such a hospital actually received Magnet status. However, interpretation of the ATO is less straightforward. That is, it requires careful elucidation of the subset of Magnet hospitals that receive non-zero weights. One strategy, outlined in Ben-Michael et al. (2024), uses descriptive statistics to compare how the treated population in the ATO changes relative to the overall treated population. As we outline below, we target the ATO in our analysis of Magnet hospitals because there is very limited overlap between treated and control hospitals, and so it is difficult to estimate the ATT. Next, we review the two primary COS designs, which rely on different identification assumptions.

2.2 Identification and estimation in COS designs

There are two primary identification strategies in COS designs, and these different identification strategies will have a direct effect on how we develop sensitivity analysis methods. Specifically,

[1] Note that although we retain the i subscript for clarity, in this setup τ^{ATT} and τ^{ATO} are population-level quantities that average over all of the units in the population.

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Ye et al. (2024) analysed identification under two alternative COS designs: the Cluster-Unit Design (CUD) and the Cluster-Only Design (COD). Both designs target the same causal estimand but rely on different identification assumptions.

In the COD, cluster assignment is fixed, or not impacted by treatment, and we only need to account for potential bias due to the assignment of clusters into treatment. On the other hand, under the CUD cluster assignment can be impacted by treatment, leading to differential selection of units into clusters: the mix of units within each cluster can change depending on which clusters are treated. Differential selection would be present in our example if sicker patients seek care at Magnet Nursing hospitals due to their Magnet status. As such, with the CUD, we need to additionally account for units differentially selecting into the clusters, which depends on the treatment assignments. Next, we review the differing identification assumptions for each design. We will focus on identifying the potential outcome under control among the treated group—i.e. μ_0, the unobserved portion of the ATT—but identification for the ATO follows similarly, with weaker overlap conditions.

Identification of μ_0 under the COD requires the following set of assumptions.

Assumption 1 ((The Cluster-Only-Design)). The following three conditions hold:

- Cluster assignments are not affected by treatment: $\boldsymbol{J}(\boldsymbol{a}) = \boldsymbol{J}$ for all $\boldsymbol{a} \in \{0, 1\}^m$.
- Ignorable treatment assignment: $A_{J_i} \perp\!\!\!\perp (\boldsymbol{J}, X_i, Y_i(a, \boldsymbol{j})) \mid K_{J_i}$ for all $a = 0, 1, \boldsymbol{j} \in \{1, \ldots, m\}^n$.
- Overlap: $\pi(k) < 1$ for all $k \in \mathcal{K}$.

For the COD, these assumptions form an ignorability assumption based only on cluster-level covariates; there is no need to condition on unit-level covariates. That is, under the COD we assume that there are no unobserved *cluster-level* covariates that are associated with the cluster-level treatment assignment and the unit-level potential outcomes. In the Magnet hospital application, this would imply that we do not need to condition on *patient-level* covariates, and selection of hospitals into Magnet treatment status is only a function of observed hospital covariates. Under this assumption, following Ye et al. (2024) we can identify μ_0 as $\mu_0 = \mathbb{E}[m(0, K_{J_i}) \mid A_{J_i} = 1]$. Note that for the ATO the overlap condition in Assumption 1 can weaken to only requiring that $0 < \pi(k) < 1$ for *some* $k \in \mathcal{K}$.

Next, identification of μ_0 under the CUD requires the following set of assumptions.

Assumption 2 ((The Cluster-Unit-Design)). The following three conditions hold:

- Cluster assignments impact outcomes only through cluster-level covariates: For two cluster assignment vectors $\boldsymbol{j}$ and $\boldsymbol{j}'$ such that the treatment assignments are the same for all units—$a_{j_i} = a_{j'_i}$ for all $i = 1, \ldots, n$—and the cluster-level covariates are the same for all clusters—$K_\ell(\boldsymbol{j}) = K_\ell(\boldsymbol{j}')$ for all $\ell = 1, \ldots, m$ i—the potential outcomes for unit i are equal $Y_i(a_{j_i}, \boldsymbol{j}) = Y_i(a_{j'_i}, \boldsymbol{j}')$. So we can denote the potential outcome as $Y_i(a, k)$, the potential outcome for unit i when treatment and cluster assignment lead to unit i being assigned to a cluster with treatment assignment a and cluster-level covariates k.
- Ignorable treatment and cluster assignment: $A_{J_i} \perp\!\!\!\perp Y_i(a, k) \mid X_i, K_{J_i}$ for all $a = 0, 1, k \in \mathcal{K}$.
- Overlap: $\pi(k, x) < 1$ for all $k \in \mathcal{K}, x \in \mathcal{X}$.

The first condition in Assumption 2 further restricts the structure of spillovers. The cluster assignment vector $\boldsymbol{j}$ only affects the potential outcomes through its impact on the cluster-level covariates $K_\ell(\boldsymbol{j})$. This is akin to a stratified interference assumption (Hudgens & Halloran, 2008): the specific identities of the units that are assigned to each cluster do not affect outcomes, only summaries and aggregates of their characteristics.

Compared to Assumption 1, for the CUD in Assumption 2, the ignorability assumption requires conditioning on both the cluster-level and individual-level covariates. Cluster-level covariates drive the assignment of clusters to treatment, and both cluster- and unit-level covariates together

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drive the selection of individuals into clusters, affecting which units ultimately receive treatment. For the Magnet hospital application, this ignorability assumption implies that we need to condition on both patient- and hospital-level covariates, and is more plausible than the ignorability assumption under the COD design. It is hard to rule out that some patients visit Magnet hospitals due to the Magnet certification and the impact certification has on the type of patients the hospital sees. Under Assumption 2, we can identify μ_0 as $\mu_0 = \mathbb{E}[m(0, K_{J_i}, X_i) \mid A_{J_i} = 1]$. Note again that for the ATO the overlap assumption is weakened to $0 < \pi(k, x) < 1$ for *some* $k \in \mathcal{K}$ and $x \in \mathcal{X}$. In sum, the key distinction between these two COS designs is the conditioning sets. Both designs target the same causal estimand μ_0, but lead to different identification formulae and so are associated with different *statistical* estimand. For the COD, we only need to condition on the cluster-level covariates $\mathcal{K}$, and the statistical estimand is $\mathbb{E}[m(0, K_{J_i}) \mid A_{J_i} = 1]$. For the CUD, we need to condition on the combined set of covariates $\mathcal{R}$, and the statistical estimand is $\mathbb{E}[m(0, K_{J_i}, X_i) \mid A_{J_i} = 1]$. In a sensitivity analysis, we seek to characterize the bias that results when an investigator omits a covariate from either $\mathcal{R}$ or $\mathcal{K}$.

Here, we consider estimation of treatment effects for COS designs using weighting estimators. First, we define $\hat{w}_i$ as a set of estimated weights. The weighting estimator has the form[2]

$$\hat{\tau}(\gamma) = \frac{1}{n_t}\sum_{i=1}^{n} A_{J_i}\hat{w}_i Y_i - \frac{1}{n_c}\sum_{i=1}^{n}(1 - A_{J_i})\hat{w}_i Y_i. \tag{3}$$

Ye et al. (2024) outlined how to estimate the weights using model-based methods via logistic regression. Ben-Michael et al. (2024) developed a balancing weighting estimator that solves a convex optimization problem to find a set of weights that directly minimize a measure of covariate imbalance subject to an additional constraint or penalty on the complexity of the weights. Both estimation methods, however, generate a set of weights that are used to estimate treatment effects in the same way, and both methods only use the cluster-level covariate conditioning set $\mathcal{K}$ under the COD and the full conditioning set $\mathcal{R}$ under the CUD. Finally, we note that although the statistical estimands under the COD and CUD use different conditioning sets, they are still aggregations of individual-level outcomes and so statistical inference for both statistical estimands require accounting for the clustered nature of the data (Ben-Michael et al., 2024, develop cluster-robust variance estimators for both statistical estimands).

To develop sensitivity analyses, we consider scenarios where the weights are estimated using either $\mathcal{R}$ or $\mathcal{K}$ when the conditioning sets actually have the following form: $\mathcal{R} = \{\mathcal{K}, \mathcal{X}, \mathbf{U}\}$ or $\{\mathcal{K}, \mathbf{U}\}$ where $\mathbf{U}$ represents omitted variables or confounders. We compare these misspecified weights to w_i^*, a set of ideal weights that would result from conditioning on $\mathcal{R} = \{\mathcal{K}, \mathcal{X}, \mathbf{U}\}$. In this scenario, estimates based on only $\mathcal{K}$ or $\mathcal{K} \times \mathcal{X}$ will display some level of bias, while estimates based on w_i^* would be unbiased and consistent.

In our discussion below, we will drop the unit subscript i and cluster subscript ℓ when it is clear from context. For variables defined at the cluster-level, such as the treatment assignment A_ℓ and cluster-level covariates K_ℓ, we will index them via a unit's cluster assignment, so that, e.g. A_J and K_J are understood to be the treatment assignment and covariates for the cluster that a unit is assigned to A_{J_i} and K_{J_i}.

3 The causal effect of magnet nursing hospitals

Given that sensitivity analyses are performed after estimating treatment effects, we next estimate the Magnet hospital treatment effect. The first step in our analysis is estimation of the weights. We assume that we are under the CUD design. We adopt the CUD design, since patients and providers may be aware of the Magnet nursing certification and this may shift patient referral patterns. Given this decision, we estimated weights that condition on both patient- and hospital-level covariates using balancing weights tailored to the COS context (Ben-Michael et al., 2024).

First, we estimated weights for the ATT estimand. We selected the hyperparameters using the data-driven procedure outlined in Ben-Michael et al. (2024). In Figure 1, we present balance statistics comparing Magnet and non-Magnet hospitals before and after weighting for the subset of

[2] Note that for the ATT, the weights on the treated unit are all equal to 1, i.e. $\hat{w}_i = 1$ for all i such that $A_{J_i} = 1$.

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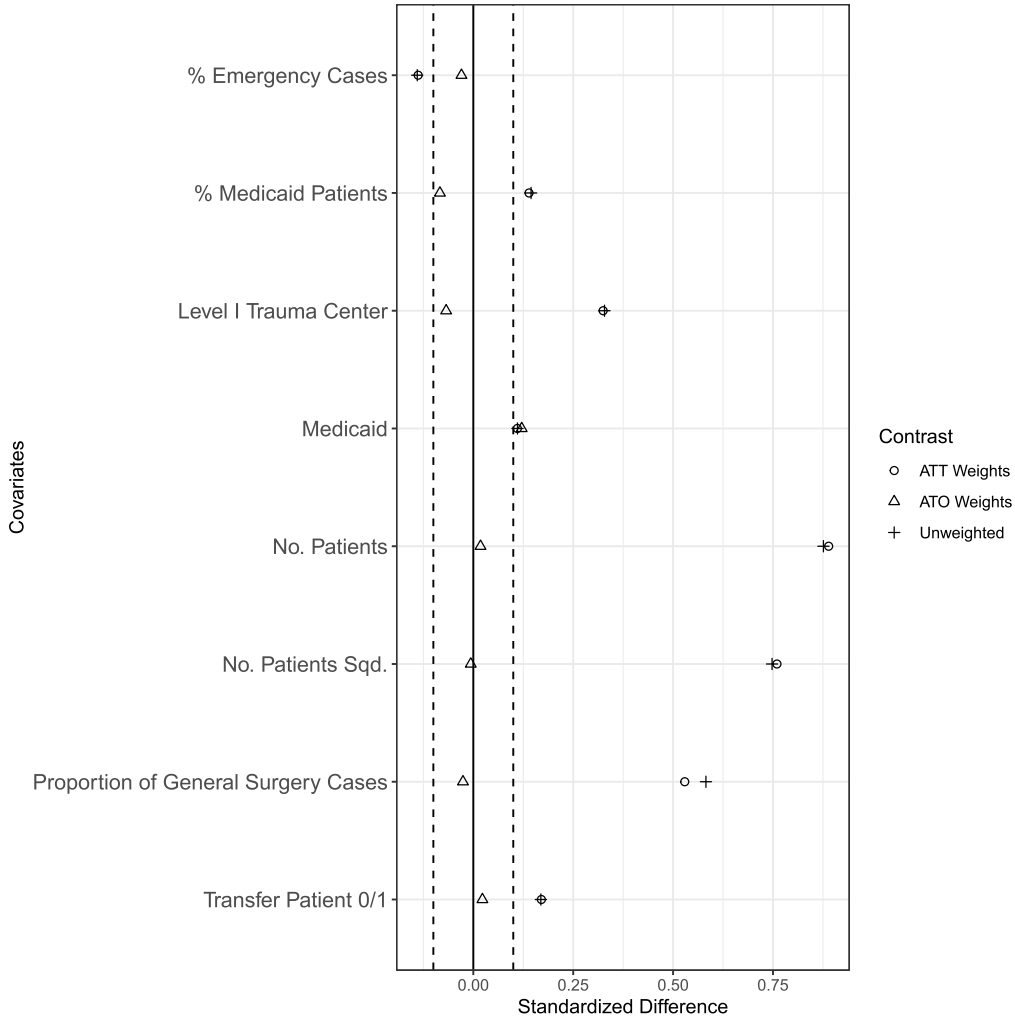

Figure 1 Plot comparing balance measured by standardized difference before and after weighting for the ATT and ATO estimands.

covariates with the largest imbalances at baseline. Notably, there are three covariates in particular that are highly imbalanced: the number of patients, the number of patients squared, and the proportion of cases that are admitted for a general surgery procedure. Notably, weights that target the ATT also fail to balance hospitals on these three covariates. This means that Magnet hospitals tend to have much larger patient volumes than non-Magnet hospitals and have a larger share of EGS patients, and there appear to be few non-Magnet hospitals that are comparable on these dimensions. Large imbalances after weighting are often indicative of a lack of overlap between treated and control clusters.

Given the lack of overlap between the treated and control clusters, we next estimated subset weights that target the average treatment effect ATO COS estimand (Ben-Michael et al., 2024). We selected the hyperparameters via the same data-driven process. Figure 1 also contains balance statistics before and after weighting for the ATO estimand. By focusing on the overlap population, we are able to estimate weights that better balance the treated and control groups (Figure 1). For the subset weights, all the standardized differences are now less than 0.10. Figure 2 contains histograms comparing the distribution of the weights for the ATT and ATO estimands. For the ATT estimand, we omit the treated weights, which are all one by design. Notably, the magnitude of the ATT weights is not overly extreme, with only one weight larger

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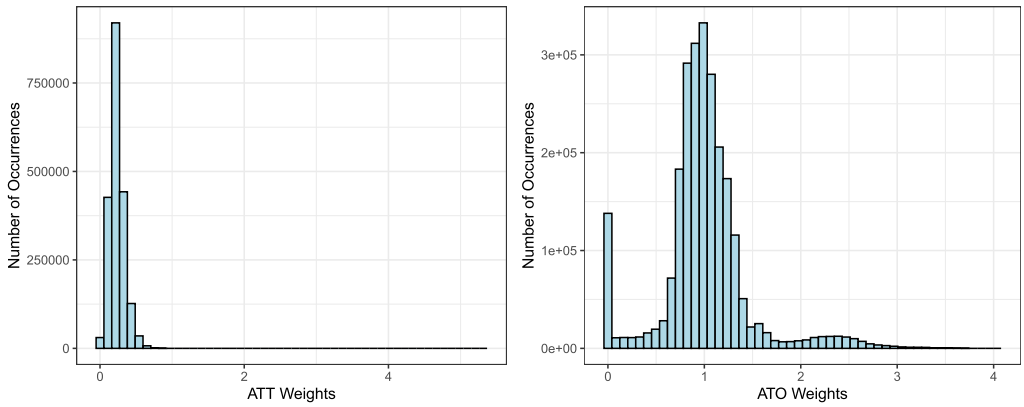

Figure 2 Histograms of the weight distributions for the ATT and ATO estimands.

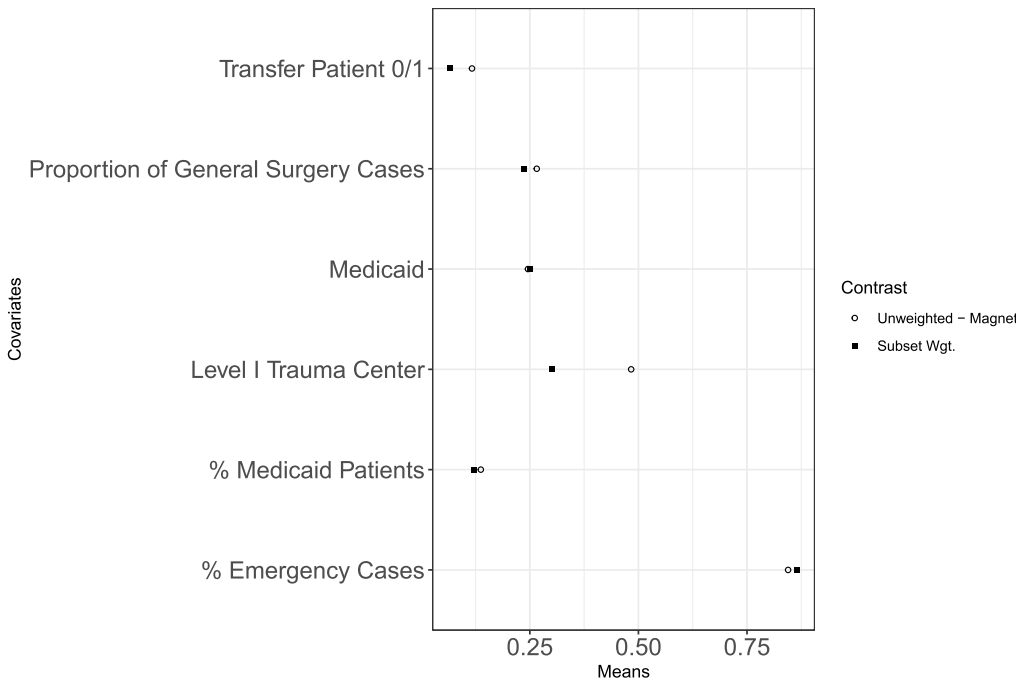

Figure 3 Magnet hospital means before and after ATO weights.

than 5. Figure 3 contains the means for Magnet Hospital before and after apply the ATO weights. This provides some context on how much the ATO estimand alters the treated group.

As such, we are able to render the treated and control groups comparable once we focus on the set of Magnet hospitals that overlap with the control hospitals. Critically, however, it is key to understand that the interpretation of this estimand is quite different. Here, this represents the effect of Magnet nursing hospitals for the subset of Magnet hospitals that overlap with the control hospitals. Table 1 contains the point estimates based on the estimators in Ben-Michael et al. (2024). For the ATT estimand, given the large imbalances, we include additional bias correction via an outcome model. For adverse events, the estimated Magnet Hospital treatment effect is 0.43% and the confidence intervals include zero. For FTR, the estimated point estimate is −1.9%, and the confidence interval also includes zero. For the ATO estimand, the estimated Magnet Hospital adverse event treatment effect is −1.7%. This estimate is statistically significantly

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Table 1 The effect of magnet hospitals on adverse events and FTR: ATT and ATO estimands.

	ATT	ATO
Adverse events	0.43	−1.7
	[−1.1, 1.9]	[−3.3, −0.1]
Failure to rescue	−1.9	−3.3
	[−3.8, 0.01]	[−5.6, −1.1]

different from zero. The FTR treatment effect is −3.3% and is also statistically significant. Thus, we see that once we adjust for confounders, under the ATO estimand, Magnet Hospitals have lower rates of adverse events compared to non-Magnet hospitals. For the ATT estimand, we observe more limited evidence that Magnet Hospitals have lower rates of FTR. Next, we develop methods to understand whether these conclusions are sensitive to the presence of unobserved confounders.

4 Bias decomposition for clustered treatment assignments

Next, we derive the sources of bias when there are omitted confounders for a clustered treatment assignment. Critically, we show that the amount of bias depends on whether the omitted confounder is at the unit or cluster level. First, we assume that we omit $\{U, V_J\}$, where U is an individual-level covariate, and V_J is a cluster-level covariate. Let $\pi(K_J, X) := \Pr(A_J = 1 \mid K_J, X)$ represent the observed population-level propensity score. Furthermore, define $\pi^*(K_J, X, V_J, U) := \Pr(A_J = 1 \mid K_J, X, V_J, U)$ as the *oracle* propensity score, since it includes the omitted covariates. We first decompose the bias that arises from omitting $\{V_J, U\}$ for the ATT estimand. We provide the same decomposition for the ATO estimand below, and for the ATE in the supplemental materials.

Throughout the paper, we focus on the large-sample limits of the inverse propensity score weights. As a result, we do not explicitly consider the estimation error that would arise in estimating the weights. Furthermore, while we formulate the bias with respect to omitted covariates, researchers can similarly apply the proposed framework to consider misspecification concerns. For example, if researchers are using a linear probability model, the omitted variable can be formulated as non-linear combinations of the observed covariates.

4.1 The ATT estimand

When the ATT is the target estimand in the CUD, the mis-specified population-level weights have the following form:

$$w^{\text{ATT}} = \frac{\pi(K_J, X)}{1 - \pi(K_J, X)}.$$

The correctly specified oracle weights, w_i^* are a function of the oracle propensity scores,

$$\begin{aligned} w^{\text{ATT}*} &= \frac{\pi^*(K_J, X, V_J, U)}{1 - \pi^*(K_J, X, V_J, U)} \\ &= \underbrace{\frac{\pi(K_J, X)}{1 - \pi(K_J, X)}}_{\equiv w_i} \cdot \underbrace{\frac{\Pr(V_J \mid K_J, X, A_J = 1)}{\Pr(V_J \mid K_J, X, A_J = 0)}}_{\text{Imbalance in Cluster-Level } V_J} \cdot \underbrace{\frac{\Pr(U \mid V_J, K_J, X, A_J = 1)}{\Pr(U \mid V_J, K_J, X, A_J = 0)}}_{\text{Imbalance in Indiv.-Level } U}. \end{aligned} \tag{4}$$

Close inspection of these weights reveals a number of key insights. First, we note that the error in the weights is driven by the residual imbalance in both U and V_J. Second, the imbalance in U (the individual-level bias) depends on how U depends on not only the *observed* covariates $\{K_J, X\}$, but also the omitted cluster-level confounder V_J. This result implies that for clustered treatment assignments, we must account for two different types of potential omitted confounders that can

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result in biased estimates, which is distinct from existing sensitivity approaches for weighting estimators.

We decompose the bias from omitting $\{V_J, U\}$ in the following theorem:

Theorem 4.1 ((Bias from Omitting Confounders)). Let the weights estimated be $w(K_J, X)$. Let $w_i^* \equiv w(K_J, X, V_J, U)$. Now let an intermediary set of weights, in which we omit U, but not V_J to be represented as $w(K_J, X, V_J)$. Then, the bias from omitting $\{V_J, U\}$ can be written as

$$\begin{aligned}\text{Bias}(\hat{\tau}_{\text{ATT}}) = \sqrt{\frac{1}{1-R_V^2}} \cdot \Bigg[& \underbrace{\text{cor}\big\{w(K_J, X) - w(K_J, X, V_J), Y \mid A_J = 0\big\}\sqrt{R_V^2}}_{\textit{Bias from cluster-level confounder}} \\ & + \underbrace{\text{cor}\big\{w(K_J, X, V_J) - w^*(K_J, X, V_J, U), Y \mid A_J = 0\big\}\sqrt{\frac{R_U^2}{1-R_U^2}}}_{\textit{Bias from unit-level confounder}} \Bigg] \\ & \times \underbrace{\sqrt{\text{var}(Y \mid A_J = 0) \cdot \text{var}(w(K_J, X \mid A_J = 0))}}_{\textit{(Estimable) Scaling Factor}},\end{aligned}$$

where $R_V^2 = \text{var}\{w(K_J, X) - w(K_J, X, V_J)\}/\text{var}\{w(K_J, X, V_J)\}$ represents the residual imbalance in the cluster-level confounder V_J and $R_U^2 = \text{var}\{w(K_J, X, V_J) - w^*(K_J, X, V_J, U)\}/\text{var}\{w^*(K_J, X, V_J, U)\}$ represents the residual imbalance in the individual-level confounder U.

Theorem 4.1 decomposes the bias into three parts: (1) the bias from omitting a cluster-level confounder, (2) the bias from omitting a unit-level confounder, and (3) an estimable scaling factor. Next, we provide some interpretation for the key components in this bias decomposition.

Bias from omitting a cluster-level confounder. First consider a setting in which we have only omitted a cluster-level confounder and so there is no bias from omitting a unit-level confounder. The resulting bias can then be written as

$$\text{cor}\big\{w(K_J, X) - w(K_J, X, V_J), Y \mid A_J = 0\big\}\sqrt{\frac{R_V^2}{1-R_V^2} \cdot \text{var}(Y \mid A_J = 0) \cdot \text{var}\big\{w(K_J, X) \mid A_J = 0\big\}}. \tag{5}$$

There are parallels between this bias decomposition and the one in standard omitted variable bias settings (e.g. Huang, 2024; Huang & Pimentel, 2025). In particular, the magnitude of the resulting bias will depend on how much residual imbalance there is in the cluster-level confounder that cannot already be explained by the observed covariates, represented by R_V^2, and a correlation term, which captures how related the imbalance is in the cluster-level confounder to the outcome process (i.e. $\text{cor}\{w(K_J, X) - w(K_J, X, V_J), Y \mid A_J = 0\}$).

The R_V^2 value is bounded between [0, 1], such that if R_V^2 is close to 1, this implies there is greater residual imbalance in the omitted cluster-level confounder. Notably, R_V^2 represents the residual imbalance in the omitted confounder that cannot be explained by the observed covariates. For example, in our Magnet Nursing study, one key concern is that we are not fully controlling for hospital quality factors that might be related to both the certification process and patient outcomes. In our analysis, we seek to make hospital comparable in terms of quality, so we controlled for observed covariates such as patient volume and trauma centre status. As such, if we believe that much of the imbalance in hospital quality can be explained by patient volume, trauma centre status and other hospital-level covariates, then the R_V^2 value will be relatively small. In contrast, if there is a large portion of the imbalance in hospital quality that cannot be explained by these observed covariates, then there will be a greater degree of residual imbalance, resulting in a relatively high R_V^2 value.

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Bias from omitting an individual-level confounder. Next, we consider the scenario in which there are no omitted cluster-level confounders, but an individual-level confounder has been omitted,

$$\text{cor}(w(K_J, X, V_J) - w^*(K_J, X, V_J, U), Y \mid A_J = 0) \cdot \sqrt{\frac{R_U^2}{1 - R_U^2}\text{var}(Y \mid A_J = 0)\text{var}(w(K_J, X) \mid A_J = 0)}.$$

Similar to Equation (5), the bias is driven by an R^2 value and a correlation term. R_U^2 represents the residual imbalance in an *individual*-level confounder. For example, in our analysis, we include covariates to capture the overall patient-mix in each hospital, and in particular try to include controls for the health status of each patient. Now, R_U^2 represents the imbalance in patient-mix and health status that cannot be explained by the observed covariates used in the analysis. If this term is large (i.e. R_U^2 is close to 1), this implies that even after accounting for the observed covariates, such as comorbidities and demographics, there is still residual imbalance in the unobservable components of the patient mix. In contrast, if $R_U^2 \approx 0$, this implies that the observed covariates explain the most of this variation.

It is worth noting that there is an asymmetry in these two bias decompositions. Notably, the bias expression is scaled by $\sqrt{\frac{1}{1-R_V^2}}$. As a result, any imbalance in an omitted cluster-level confounder will *amplify* the bias incurred from omitting an individual-level confounder. Consider a setting in which the omitted cluster-level confounder is highly imbalanced (i.e. R_V^2 is close to 1), but is only weakly related to the outcome (i.e. $\text{cor}\{w(K_J, X) - w(K_J, X, V_J), Y \mid A_J = 0)$ is close to zero). Then, even if the bias from omitting a cluster-level confounder is relatively small, *any* bias from the unit-level confounder will then be amplified because of the $\sqrt{\frac{1}{1-R_V^2}}$ factor. This same logic applies to observed covariates. That is, small imbalances in hospital-level covariates will amplify the bias from imbalances in patient-level covariates. As such, analysts should prioritize balance on the cluster-level covariates when estimating the weights. This point also points to one reason investigators prefer the ATO estimand to the ATT estimand. In our Magnet Nursing analysis, we found it difficult to balance the hospital-level covariates when we targeted the ATT. The residual imbalance in hospital-level covariates for the ATT estimand may amplify the bias from the imbalance in the patient covariates. However, under the ATO estimand, we are able to achieve far better balance on hospital covariates, which will avoid this kind of bias amplification.

Scaling Factor. The final piece of the bias decomposition is $\sqrt{\text{var}(Y \mid A_J = 0) \cdot \text{var}(w(K_J, X) \mid A_J = 0)}$, which acts as a scaling factor that amplifies the effect of omitted confounders. To provide some intuition, we consider the setting in which $\text{var}(Y \mid A_J = 0)$ is large. In this case, there is a high degree of heterogeneity in the control group outcomes that is confounded with the biased treatment assignment process. Similarly, as $\text{var}(w(K_J, X) \mid A_J = 0)$ increases, there is greater heterogeneity in the observed covariates, K_J and X, for the control group. The presence of such heterogeneity may make it more difficult to balance cluster- and individual-level covariates, increasing the potential for bias.

In Theorem 4.1, we choose to parameterize the error from omitting $\{U, V_J\}$ by first considering the imbalance in the cluster-level omitted confounder V_J, and then considering the residual imbalance in the individual-level omitted confounder U after controlling for $\{V_J, K_J, X\}$. This parameterization is not unique—researchers could equivalently choose to parameterize the error in the weights from omitting $\{V_J, U\}$ by considering first the imbalance in the individual-level U, and then looking at the residual imbalance in V_J, after accounting for $\{U, K_J, X\}$. However, our choice of parameterization maps intuitively to how we differentiate between the COD and CUD designs. In particular, in a COD design, the imbalance in individual-level U term in Equation (4) is exactly equal to 1, as we do not need to condition on individual-level covariates.[3]

[3] If we chose to first condition on the individual-level U, we would still observe a cross dependency in the bias from the cluster-level confounder and the bias from the unit-level confounder. In particular, we can re-write the error term to be a function of the estimated weights w_i, the imbalance in U, and then the *residual* imbalance in the cluster-level omitted confounder.

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4.2 The ATO estimand

When the ATO is the target estimand in the CUD, the mis-specified population-level weights have the following form:

$$w^{\text{ATO}} = \begin{cases} \pi(K_J, X), & \text{if } A_J = 0, \\ 1 - \pi(K_J, X), & \text{if } A_J = 1. \end{cases}$$

The correctly specified oracle weights, w_i^* are a function of the oracle propensity scores,

$$\begin{aligned} w^{\text{ATO}*} &= \begin{cases} \pi(K_J, X, V_J, U), & \text{if } A_J = 0, \\ 1 - \pi(K_J, X, V_J, U), & \text{if } A_J = 1, \end{cases} \\ &= \begin{cases} \pi(K_J, X) \cdot \dfrac{P(V_J \mid K_J, X, A_J = 1)}{P(V_J \mid K_J, X)} \cdot \dfrac{P(U \mid K_J, X, V_J, A_J = 1)}{P(U \mid K_J, X, V_J)}, & \text{if } A_J = 0, \\ 1 - \pi(K_J, X) \cdot \dfrac{P(V_J \mid K_J, X, A_J = 1)}{P(V_J \mid K_J, X)} \cdot \dfrac{P(U \mid K_J, X, V_J, A_J = 1)}{P(U \mid K_J, X, V_J)}, & \text{if } A_J = 1. \end{cases} \end{aligned} \tag{6}$$

Equation (6) highlights that we can similarly decompose the oracle ATO weights as a function of the estimated ATO weights w_i^{ATO} and residual imbalance in the cluster-level omitted confounder, as well as the residual imbalance in the individual-level omitted confounder.

Next, we require the bias decomposition for the ATO estimand. We include the full bias decomposition in online supplementary material, Appendix B. In that decomposition, we show that the bias for the ATO from omitting $\{V_J, U\}$ decomposes into (1) correlation terms that measure how related the imbalance in the omitted confounders are to the outcomes across treatment and control, (2) how much residual imbalance there is in the cluster-level omitted confounder and the individual-level omitted confounder, and (3) a scaling factor that depends on how much variance there is in the outcomes and estimated weights w_i^{ATO}. One advantage of the ATO relative to the ATT (or the ATE) is that the scaling factor in the bias decomposition depends on the variance of the estimated weights. However, since the weights for the ATO are just the propensity scores themselves, they will have lower variance than the weights for the ATT, which are the inverse propensity scores. As a result, the bias from omitting a confounder for the ATO may be smaller than the bias from omitting a similar confounder in the ATT setting.

5 Sensitivity models

To assess the sensitivity to omitted confounders, we introduce two different weighted sensitivity models for the COS setting. Following Huang et al. (2025), we define a *sensitivity model* f_v as a set of hypothetical, ideal weights w^*, which is constructed by constraining a local neighbourhood around the estimated weights w. The size of the neighbourhood around w is controlled by a parameter. The larger the parameter, the more we are allowing the ideal weights to differ from the estimated, observed weights.[4] Thus, as the parameter gets larger, we are allowing for increasing amounts of unobserved confounding. Below, we show that for the COD setting, sensitivity analysis models developed for the unclustered treatment assignment setting can be adapted to COS applications in a straightforward fashion. However, for the CUD, we derive new sensitivity analysis models that allow researchers to consider unit- or cluster-level confounders, or both.

Formally, we define a sensitivity model for $\{V_J, U\}$ as $v(\Gamma, w(K_J, X))$, where

$$v(\Gamma, w(K_J, X)) := \left\{ w^*(K_J, X, \{V_J, U\}) : f_v(w(K_J, X), w(K_J, X, \{V_J, U\})) \leq \Gamma \right\}.$$

[4] Traditionally, the sensitivity models have focused on constraining errors in the underlying propensity scores. In the following paper, we focus on constraining the error in the *weights*.

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In words, this sensitivity model defines the set of possible oracle propensity scores $\pi^*(K_J, X, \{V_J, U\})$ that could exist, given a constraint on how different the oracle propensity scores are to the estimated propensity scores. The choice of f_v encodes how researchers are choosing to quantify the dissimilarity between the estimated propensity scores and the hypothetical, oracle propensity scores. Different choices of dissimilarity measures correspond to different sensitivity models in the literature. For any given sensitivity model, there is a sensitivity parameter—represented by Γ—which constraints the total amount dissimilarity the sensitivity model allows for.

Given a sensitivity model $f_v(\Gamma, w(K_J, X))$, researchers can then optimize over the uncertainty set to obtain the partially identified region,

$$\left[\inf_{\tilde{w}\in v(\Gamma,w)} \tau(\tilde{w}), \sup_{\tilde{w}\in v(\Gamma,w)} \tau(\tilde{w})\right].$$

This partially identified set represents the upper and lower bounds on the treatment effect given the sensitivity model. The minimum value of Γ represents the case of no unobserved confounders, and partially identified set will be point identified. Researchers can then increase the values of Γ to obtain larger partially identified sets. Investigators are typically interested in finding the value of Γ that produces a partially identified region that contains zero. When small values of Γ produce a partially identified set that includes zero, this indicates the results sensitive to bias from an unobserved confounder. In the following two subsections, we outline two different sensitivity models for COS designs. The first is the *marginal sensitivity model*, which constrains the worst-case error across the weights, while the second is the *variance sensitivity model*, which constrains the weighted average error.

5.1 Marginal sensitivity model

We first consider the marginal sensitivity model from Tan (2006). The marginal sensitivity model constrains the worst-case error between the ideal weights and the estimated weights,

$$v_{\text{MSM}}(\Lambda, w) = \left\{w^* : \Lambda^{-1} \leq \frac{w(X, K_J)}{w^*(X, K_J, V_J, U)} \leq \Lambda, \quad \text{for } x \in \mathcal{X}, u \in \mathcal{U}, k_J \in \mathcal{K}_J, v \in \mathcal{V}_J\right\},$$

where $\Lambda \geq 1$. In other words, the distance measure used to constrain w and w^* is an L_∞ norm. In the COS design, the marginal sensitivity model from Zhao et al. (2019), which assumes that treatment assignment is unclustered, can be directly applied. In the CUD setting, it can be helpful to consider an amplification of the parameter Λ (see Section 7 for details).

For a fixed Λ value, researchers can optimize over the set $v_{\text{MSM}}(\Lambda, w)$ to find the maximum and minimum values the estimated treatment effect can take on. In practice, the optimization procedure amounts to solving a linear programming problem (Zhao et al., 2019). Recent literature has also introduced alternative estimation approaches that allow researchers to construct sharp bounds under the marginal sensitivity model by using a quantile regression (e.g. Dorn & Guo, 2023).

5.2 Variance sensitivity model

Bounds based on a marginal sensitivity model tend to be overly conservative in that they result in a partially identified set that is very wide (Huang & Pimentel, 2025; Huang et al., 2025). Next, we develop sensitivity methods using a variance sensitivity model. Unlike the marginal sensitivity model, which constrains the worst-case error that arises in the weights, the variance-based sensitivity model constrains a weighted average error. More formally, it will constrain the distributional difference between the hypothetical ideal weights and the estimated weights,

$$v_{vbm} := \left\{w^* : 1 \leq \frac{\text{var}(w(X, K_J, V_J, U))}{\text{var}(w(X, K_J))} \leq \frac{1}{1 - R^2}\right\}.$$

An advantage to the variance-based approach is that the constraint on the difference between the variance in the ideal weights and the variance of the estimated weights can be parameterized with

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respect to an R^2 value. We can view the R^2 value as how much unexplained variance there is leftover in the ideal weights that cannot be explained by the estimated weights. Huang and Pimentel (2025) showed that in settings where treatment assignment is unclustered that bounds from a variance sensitivity model were less pessimistic. We will show that using a variance sensitivity model in the COS setting is less pessimistic than the methods based on the marginal sensitivity model.

Another advantage is that we can calculate a closed-form representation of the maximum and minimum bias. For the ATT under the CUD, the maximum bias for a specified R^2 value can be written as follows:

$$\max_{\tilde{w}\in v_{\text{vbm}}(R^2)} \left|\text{Bias}(\hat{\tau}_{\text{ATT}})\right| \leq \sqrt{1-\text{cor}^2\{w(X, K_J), Y \mid A_J = 0\}} \times \sqrt{\frac{R^2}{1-R^2}\text{var}\{w(X, K_J) \mid A_J = 0\} \cdot \text{var}(Y \mid A_J = 0)}.$$

This leads to a similar decomposition as the bias decomposition from Theorem 4.1, but bounds the maximum correlation between the imbalance between the estimated and oracle weights and the outcomes. The upper bound is a function of the correlation between the estimated weights $w(X, K_J)$ and the outcomes Y. Informally, in settings when *observed* covariates $\{X, K_J\}$ are very related to the outcomes Y, then this restricts the amount of residual confounding that can exist from omitted variables $\{U, V_J\}$. This emphasizes a point made in Huang (2024), which highlights the importance of weighting on variables that are not only prognostic of the biased selection procedure A_J, but also the outcome.

Next, we derive the bound for the maximum bias for a specified R^2 value in the ATO setting,

$$\max_{\tilde{w}\in v_{\text{vbm}}(R^2)} \left|\text{Bias}(\hat{\tau}_{\text{ATO}})\right| \leq \overbrace{\sqrt{1-\text{cor}^2\{w(X, K_J), Y \mid A_J = 1\}} \times \sqrt{\frac{R^2}{1-R^2}\text{var}\{w(X, K_J) \mid A_J = 1\} \cdot \text{var}(Y \mid A_J = 1)}}^{\text{Maximum bias across treatment units}} + \underbrace{\sqrt{1-\text{cor}^2\{w(X, K_J), Y \mid A_J = 0\}} \times \sqrt{\frac{R^2}{1-R^2}\text{var}\{w(X, K_J) \mid A_J = 0\} \cdot \text{var}(Y \mid A_J = 0)}}_{\text{Maximum bias across control units}}.$$

The key difference in this bound for the ATO is that we are assuming, for the treatment units, that the error from omitting the confounders is maximally *positively* correlated with the outcomes across treatment. In contrast, for the control units, the error from omitting the confounders will be maximally *negatively* correlated with the outcomes across control. This implies that we are mistakenly overweighting units with larger outcome values in the treatment group, while underweighting units with larger outcome values in the control group. To our knowledge, this the first time such bounds have been extended to the ATO estimand.

5.3 Inference

We conclude with a discussion of inferential methods. Both methods provide upper and lower bounds on the true treatment effect given a hypothesized level of confounding. These bounds do not, however, include any statement of statistical uncertainty. As such, inferential methods are needed to provide confidence intervals for the bounds. In settings with unit-level treatment assignment, the bootstrap is used for inference (Zhao et al., 2019). The bootstrap should also provide a method for inference in our setting as well. However, given the clustered nature of treatment assignment, it is necessarily to use the block bootstrap, which resamples the clusters with replacement (Davison & Hinkley, 1997). Below, we evaluate the use of the block bootstrap using simulation.

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6 Simulation study

6.1 Simulation 1

Purpose. Next, we conduct two simulation studies to evaluate our proposed methods. We base our simulation study on a design from Ben-Michael et al. (2024), developed to evaluate COS balancing weight methods. First, we outline the data generating process (DGP). The DGP is built from real data designed to evaluate a summer school reading intervention. In the original data, there are 18 treated schools with 1,367 students, and 26 control schools with 2,060 students. The primary outcome in the data is a reading score collected after the summer school reading programme. The 5 student-level variables are previous reading and math test scores and indicator variables for race, ethnicity, and sex. The school-level covariates consist of the percentage of students who receive free/reduced price lunch, the percentage who are English language learners, the percentage of students that are proficient in math and in reading, average school-level math scores, average school-level reading scores, the share of teachers who are novice (e.g. in their first year), the rate of year-to-year staff turnover, and student average daily attendance.

Data generating process. We used this real data for the simulation in several ways. First, we regressed the observed treatment indicator on the school-level variables. Using this model, we generated school-level propensity scores which we denote as $\hat{e}(w)$. Next, we define the latent probability of treatment as

$$Z^* = (\hat{e}(w)/c) + \text{Unif}(-.5, .5),$$

where c controls the level of overlap between treated and control clusters. We then generated treatment assignments via: $Z_j = 1(Z^* > 0.25)$, where 1 is the indicator function. Next, we generated potential outcomes under control as

$$y_0 = \hat{\beta}_0 + 2.5R_{ij} + 2.5M_{ij} + 1.9P_j + \nu_1.$$

In this model, R_{ij} is student-level reading scores, M_{ij} is student-level math scores, and P_j is the percentage of students proficient in math and reading in each school. $\hat{\beta}_0$ is the intercept from an outcome model where we regressed the outcome on the student-level covariates with the basis expanded to include interactions between race and past test scores. Finally, ν_1 is a draw from a normal distribution that is mean zero with a standard deviation of 12. Next, we generated potential outcomes under treatment as $y_1 = y_0 + \tau$, and we generated simulated outcomes as $\tilde{Y}_{ij} = Z_j y_1 + (1 - Z_j)y_0$. Here, τ is the true treatment effect estimate in the simulation, and we set it to be 0.3 of a standard deviation of the raw outcome measure. Note that the average ICC across simulations was 0.29 with a standard deviation of 0.04. Finally, in terms of assigning students to clusters, we simply retain the student—school pairing that occurs naturally in the data.

Methods. Using this DGP, we conduct a simulation study to understand the performance of our proposed methods and to compare the marginal sensitivity model to the variance sensitivity model. We define w^* as set of weights calculated from the true propensity scores which are based on correctly specified models from the DGP. We consider two settings. First, we consider a setting in which the propensity scores are misspecified with respect to unit- (student) level propensity scores. In this case, we omit both R_{ij} and M_{ij} from the estimation of the propensity scores. We then define these weights as w_u. Based on w_u, we define the oracle sensitivity parameters in this setting as

$$R^2 = 1 - \frac{\text{var}(w_u)}{\text{var}(w^*)}, \quad \text{and} \quad \Lambda = \max\left\{\frac{w_u}{w^*}, \frac{w^*}{w_u}\right\}.$$

Second, we consider a setting when the propensity scores are misspecified from omitting a cluster-level covariate. We define w_v as the set of weights where the school-level test scores

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covariates are omitted from the model. Then, the oracle sensitivity parameters in this setting are defined as:

$$R^2 = 1 - \frac{\text{var}(w_v)}{\text{var}(w^*)}, \quad \Lambda = \max\left\{\frac{w_v}{w^*}, \frac{w^*}{w_v}\right\}.$$

We use this DGP to conduct two different simulation studies. In the first study, we evaluate whether our proposed sensitivity analysis methods result in valid bounds that contain the true ATT. In the second study, we evaluate the performance of the clustered bootstrap when used for inference. In all the simulations, we focus on the ATT estimand and the corresponding sensitivity analysis for this estimand.

Our first simulation study is based on the following set of steps: (1) generate data under the DGP, (2) estimate two sets of underspecified COS balancing weights—one with unit variables omitted and one with the cluster variables omitted, (3) compute the sensitivity bounds, and (4) record whether the true treatment effect falls within the sensitivity bounds and the length of the sensitivity bound. In the fourth step, we estimated four different sets of bounds, applying both the marginal- and variance-based bounds for the weights misspecified with respect to student-level covariates and school-level covariates using the oracle sensitivity parameters. In sum, we seek to understand the behavior of the sensitivity analysis when the COS balancing weights are intentionally misspecified, testing whether the sensitivity bounds indeed include the true treatment effect at the oracle values of the sensitivity parameters. We also compare the length of the two sets of bounds to understand the extent to which the marginal model is more conservative. In the simulation, we vary the overlap parameter c, using values of 1, 5, and 10 which represent poor, moderate, and good overlap, respectively. We use 1000 simulations for each value of c.

Above, we noted that the bootstrap is one way to conduct inference for either type of bounds. In a second simulation, we evaluated the use of the bootstrap as an inferential method for the sensitivity bounds. For each draw from the DGP, we estimate both the marginal- and variance-based bounds at the oracle values for the sensitivity parameter. We then re-sample from the data from this draw from the DGP, and we re-estimate both sets of bounds B times. We then use the percentiles from the bootstrap distributions as confidence intervals for the bounds. Given the clustered nature of the data, we use the block bootstrap which resamples the clusters with replacement but then retains all the units within the resampled clusters without further resampling (Davison & Hinkley, 1997). As an outcome metric, we recorded empirical coverage of the true treatment effect using the bootstrap confidence intervals. In this simulation, we used 500 simulation replications, and for each of these replications, we resampled 1,000 times for the block bootstrap.

Results. We first report the results for when the cluster-level covariates are omitted from the estimated weights in Figure 4. First, we find that the bounds based on both the variance and marginal sensitivity model include the true treatment effect 100% of the time across all three levels of overlap. However, there are stark differences in terms of the length of the bounds. First, we observe that the sensitivity bounds based on the marginal sensitivity model are always wider than the bounds based on a variance-based sensitivity model. The length of the bounds for the marginal sensitivity model is roughly constant across levels of overlap. However, the bounds for the variance-based sensitivity model become much shorter as overlap improves. However, even when the bounds from the variance-based sensitivity model are at their longest, they are still 25% shorter than those from the marginal sensitivity model. The bootstrap confidence intervals also covered the true treatment effect 100% of the time.

Next, we report the results for when the unit-level covariates are omitted from the estimated weights in Figure 5. In terms of the length of the bounds, again those from the marginal sensitivity model are much wider and do not narrow as the degree of overlap improves. Again, we observe that the variance-based sensitivity model produces bounds that are much short. Here, we find that the marginal sensitivity model bounds cover the true treatment effect 100% of the time. However, for the marginal sensitivity model, the bounds include the true value approximately 98% of the time across the three overlap conditions. As such, the less conservative bounds length

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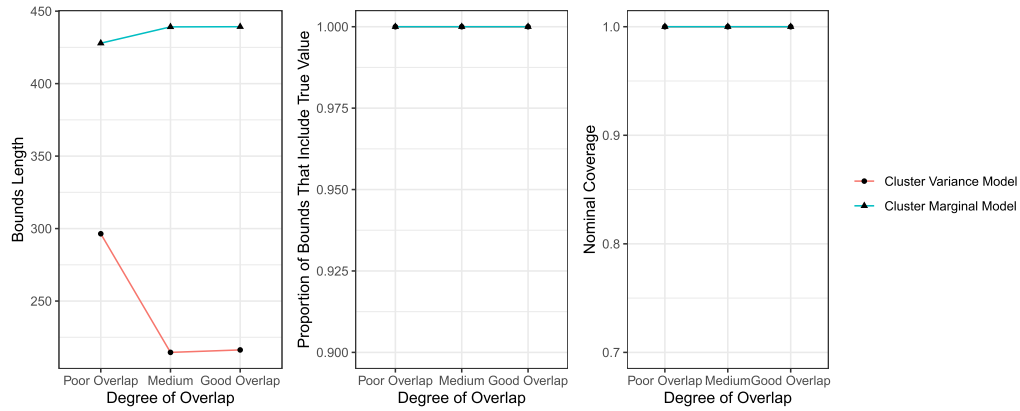

Figure 4 Length of the sensitivity bounds, proportion of times that the sensitivity bounds include true value, nominal coverage for three different levels of covariate overlap when cluster-level variables are omitted.

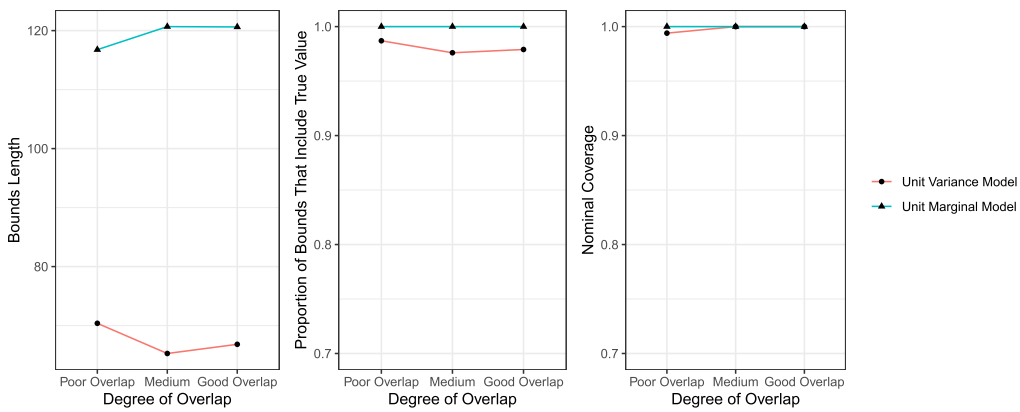

Figure 5 Proportion of times sensitivity bounds include true value and length of sensitivity bounds for three different levels of covariate overlap when unit-level variables are omitted.

does imply that sometimes the bounds do not cover the true treatment effect value. That is, under the variance-based sensitivity model, some amount of estimation error does result in some portion of the estimated bounds not including the true effect. However, the bootstrap confidence intervals also covered the true treatment effect 100% of the time.

6.2 Simulation 2

Purpose. Next, we conduct a second simulation using a DGP from Ye et al. (2024). This DGP includes both differential selection—units selecting clusters based on cluster treatment status—and stratified inference such that outcomes within clusters may affect other outcomes within the same cluster.

Data generating process. In this DGP, we first generated the baseline covariates at the cluster-level $W_{j1}, W_{j2}, j = 1, \ldots, m$ from mean zero multivariate normal distribution, with $\text{var}(W_{j1}) = 1$, $\text{cov}(W_{j1}, W_{j2}) = 0.5$ and $\text{var}(W_{j1}) = 1.25$. Next, there are five observed covariates at the unit-level in the population $X_1, \ldots, X_5$. We generate these baseline covariates as $X_{i1} \sim N(0, 1)$, $X_{i2} \sim \text{Binom}(0.4)$, and X_{i3}, X_{i4}, X_{i5} are drawn from a mean zero multivariate normal distribution with the following variance–covariance matrix,

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$$\begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & -0.5 \\ -1 & -0.5 & 1 \end{bmatrix}.$$

We use these baseline covariates to govern how units are assigned to clusters. Here, we focus on the setting where units respond to cluster treatment assignment status. Specifically, we assign the treatment according to $\text{logit}\{P(Z_j = 1 \mid W_j)\} = 0.2W_{j1} + 0.1W_{j2}$, where the model that assigns units to clusters is generated from the following model:

$$P(J_i = j \mid X_i, \boldsymbol{Z}, \boldsymbol{W}) = \frac{\exp\{(0.1W_{j1} + 0.1W_{j2}) \cdot (1 + X_{i1} + X_{i2} + X_{i3} + X_{i4} + X_{i5}) + (0.2Z_j) \cdot (1 + X_{i1} + X_{i2})\}}{\left[\sum_{j=1}^{m} \exp\{(0.1W_{j1} + 0.1W_{j2}) \cdot (1 + X_{i1} + X_{i2} + X_{i3} + X_{i4} + X_{i5}) + (0.2Z_j) \cdot (1 + X_{i1} + X_{i2})\}\right]}.$$

Here, unit to cluster assignment depends on treatment assignment. Next, we generated outcomes using the same model,

$$\begin{aligned} Y_i = {} & X_{i1} + X_{i2} + X_{i3} + X_{i4} + X_{i5} + 0.4Z_{J_i}(X_{i1} + X_{i2}) \\ & + 0.5(W_{J_i1} + h_{J_i1} + h_{J_i2} + h_{J_i3} + h_{J_i4}) + 0.1e_{J_i} + \epsilon_i, \end{aligned}$$

where $e_j, \epsilon_i \sim N(0, 1)$ for $j = 1, \ldots, m$, $i = 1, \ldots, n$. The true average treatment effect is $0.4E[X_{i1} + X_{i2}] = 0.16$.

Methods. With this new DGP, we conducted a study that follows the same template as the first simulation. That is, we again (1) generate data under the DGP, (2) estimate two sets of underspecified COS balancing weights—one with unit variables omitted and one with the cluster variables omitted, (3) compute the sensitivity bounds, and (4) record whether the true treatment effect falls within the sensitivity bounds and the length of the sensitivity bound. As before, we estimated four different sets of bounds, applying both the marginal- and variance-based bounds for the weights misspecified with respect to unit-level covariates and cluster-level covariates using the oracle sensitivity parameters. In this simulation, we focus on more than one estimand. Specifically, we estimated weights for both the ATT and ATO. Finally, we again conducted a separate simulation, where we used the clustered bootstrap to estimate confidence intervals and coverage.

Results. Figure 6 contains the results for the scenario when the cluster-level covariates are underspecified. As in the first simulation, we find that sensitivity bounds always include the true

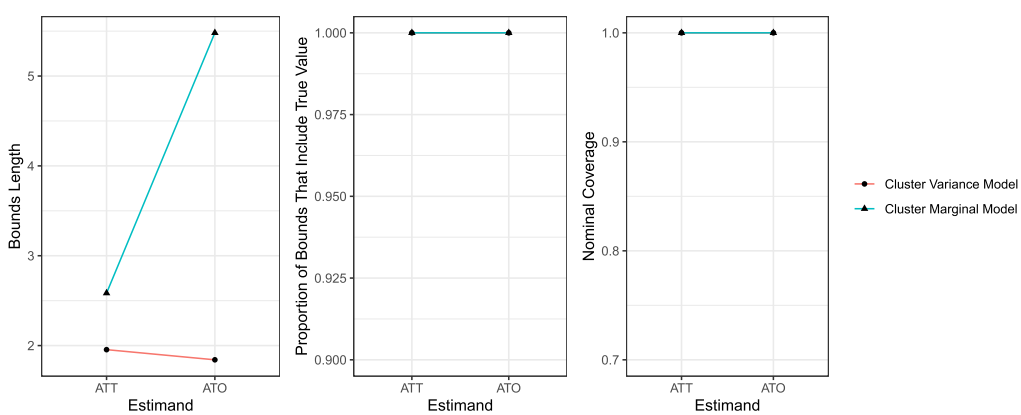

Figure 6 Length of the sensitivity bounds, proportion of times that the sensitivity bounds include true value, nominal coverage for three different levels of covariate overlap when cluster-level variables are omitted.

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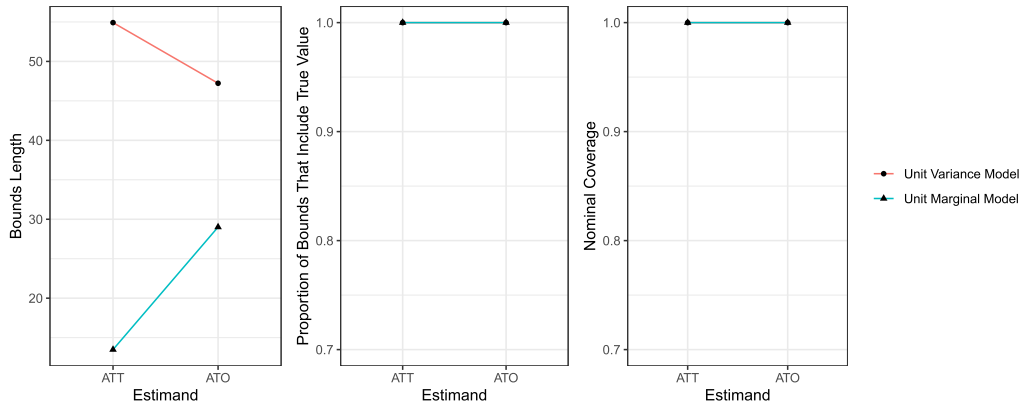

Figure 7 Proportion of times sensitivity bounds include true value and length of sensitivity bounds for three different levels of covariate overlap when unit-level variables are omitted.

treatment effect. In addition, the confidence intervals overcover in an identical fashion. As before, the variance-based sensitivity model produces sensitivity bounds that are not as wide as those based on the marginal sensitivity model. However, we now observe that the bounds from the marginal sensitivity model are much less conservative for the ATT estimand than the ATO. For the variance-based sensitivity model, the length of the bounds essentially does not vary with the estimand. Why might the length of the bounds vary with the estimand under the marginal sensitivity model? For the ATO, the bounds depend on the variance of the weights. Here, as the variance of the weights increases under the ATO, it results in much wider bounds.

Figure 7 contains the results from the scenario when the unit-level covariates are underspecified. As in all the other scenarios, we find that the performance of the bounds is validated. That is, both sets of bounds contain the true treatment effect and the confidence interval behaviour is the same. However, unlike the other settings, we see that the variance-based sensitivity model produces wider bounds than the marginal sensitivity model. This is due to the fact that in this simulation scenario, the oracle sensitivity parameters are very large for both methods of sensitivity analysis, as the unit-level covariates have a strong effects on assignment in this DGP. However, the marginal sensitivity model constructs bounds that are sample bounded, which means that above a given threshold, the estimated bounds will converge towards the range of observed outcomes. In contrast, the variance-based sensitivity model is not inherently sample bounded and will estimate bounds that continue to widen with the sensitivity parameter. We refer readers to Huang and Pimentel (2025) for more discussion.

7 Sensitivity analysis tools

Next, we develop two additional tools that can help analysts better understand the results from a sensitivity analysis. These two tools are known as amplification and benchmarking and are widely used in settings with unclustered treatment assignment (Huang et al., 2025). Here, we develop these methods for the COS design. We then apply them to the Magnet Nursing data.

7.1 Amplification

Thus far, we have considered the aggregate impact of omitting either a cluster-level or individual-level confounder. In practice, it can be helpful to consider the impact of omitting a cluster-level confounder and an individual-level confounder separately. Next, we outline this type of disaggregation, which is called an amplification. Following our earlier results, we outline amplification for both the marginal- and variance-based sensitivity analyses.

First, for the marginal sensitivity model, following Equation (4), we can write the multiplicative error between w and w^* as a product of two imbalance terms: the imbalance in the cluster-level V_J and the imbalance in the individual-level U.

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Proposition 7.1 ((Amplification of Λ)). Define Λ_V and Λ_U as

$$\Lambda_V^{-1} \leq \frac{w(X, K_J)}{w(X, K_J, V_J)} \leq \Lambda_V \quad \text{and} \quad \Lambda_U^{-1} \leq \frac{w(X, K_J, V_J)}{w(X, K_J, V_J, U)} \leq \Lambda_U,$$

for $x \in \mathcal{X}$, $u \in \mathcal{U}$, $k_J \in \mathcal{K}_J$, $v \in \mathcal{V}_J$. Then, we can amplify Λ into the product of Λ_V and Λ_U,

$$\Lambda := \Lambda_V \cdot \Lambda_U.$$

Λ_V constrains the maximum imbalance in the cluster-level confounder, while Λ_U constrains the maximum imbalance in the individual-level confounder.

Next, for the variance-based sensitivity model, we can amplify the R^2 value into two components that control the imbalance in the omitted cluster-level confounder and the imbalance in the individual-level confounder:

Proposition 7.2 ((Amplifying R^2)). Define R_V^2 and R_U^2 as follows:

$$\frac{\text{var}(w(X, K_J) \mid A = 0)}{\text{var}(w(X, K_J, V_J) \mid A = 0)} \leq 1 - R_V^2, \quad \text{and}$$
$$\frac{\text{var}(w(X, K_J, V_J) \mid A = 0)}{\text{var}(w(X, K_J, V_J, U) \mid A = 0)} \leq 1 - R_U^2,$$

for $x \in \mathcal{X}$, $u \in \mathcal{U}$, $k_J \in \mathcal{K}_J$, $v \in \mathcal{V}_J$. Then, we can amplify R^2 as

$$R^2 = 1 - (1 - R_V^2) \cdot (1 - R_U^2).$$

The amplification of the sensitivity parameter into two components allow researchers to explore the effects of cluster-level and individual-level confounders separately. This is helpful in settings where researchers have a strong substantive priors about the types of confounders they are most worried about. For example, if they are more concerned with potentially omitting a cluster-level confounder (as opposed to an individual-level confounder), the amplification allows them to disentangle the two from one another. Furthermore, amplification allows researchers to evaluate how the resulting bias changes as the cluster-level confounder (or the individual-level confounder) varies in strength (relative to the other). Finally, amplification is useful in concert with the benchmarking procedure we outline next (see Section 7.2), which we can use to calibrate potential sensitivity parameter values.

7.2 Benchmarking

Researchers can use a benchmarking analysis to calibrate sensitivity parameter values. Specifically, benchmarking allows researchers to compare sensitivity parameters associated with an omitted confounder to the confounding strength of observed covariates. Such an analysis proceeds by sequentially omitting cluster-level covariates and re-estimating the weights. A similar exercise can be done for the individual-level covariates. Researchers can additionally omit subsets of observed covariates (i.e. an individual-level covariate, and a cluster-level covariate) to estimate the bias that would occur from omitting different types of confounders. In each case, one can compare these under-specified weights to the fully specified weights to understand the effect of each type of under-specification.

To formalize benchmarking, we first define what we mean for an omitted confounder to have equivalent confounding strength to an observed covariate. For the variance-based sensitivity model, this means that the distributional difference that arises in the estimated and oracle weights is equivalent to the distributional difference that arises in the weights with and without the benchmarked covariates. More formally, let $\{X^{(b)}, K_J^{(b)}\} \in \{X, K_J\}$ represent the set of individual-level and cluster-level covariates we are trying to benchmark. Then, defining $\tilde{X} = X \backslash X^{(b)}$ and $\tilde{K}_J \backslash K_J^{(b)}$,

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a set of omitted confounders $\{U, V_J\}$ has equivalent confounding strength to the observed covariates $\{X^{(b)}, K_J^{(b)}\}$ if

$$\frac{\mathrm{var}(w(X, K_J, U, V_J) \mid A = 0) - \mathrm{var}(w(X, K_J) \mid A = 0)}{\mathrm{var}(w(X, K_J, U, V_J) \mid A = 0)} = \frac{\mathrm{var}(w(X, K_J) \mid A = 0) - \mathrm{var}(w(\tilde{X}, \tilde{K}_J) \mid A = 0)}{\mathrm{var}(w(X, K_J) \mid A = 0)}.$$

Then, we can compute the corresponding R^2 parameter for a set of omitted confounders with equivalent confounding strength as $\{X^{(b)}, K_J^{(b)}\}$,

$$R^2_{(b)} = \frac{\hat{R}^2_{-(b)}}{1 + \hat{R}^2_{-(b)}}, \quad \text{where} \quad \hat{R}^2_{-(b)} := 1 - \frac{\mathrm{var}\{w(\tilde{X}, \tilde{K}_J) \mid A = 0\}}{\mathrm{var}\{w(X, K_J) \mid A = 0\}}.$$

In settings where researchers restrict the set of benchmarked covariates to only individual-level covariates or only cluster-level covariates, then we can benchmark either the R^2_U or R^2_V value as appropriate.

For the marginal sensitivity model, benchmarking requires assuming that omitting a confounder results in the same worst-case individual-level error as omitting an observed covariate. As such, researchers can find the corresponding Λ parameter for an omitted confounder that results in the same maximum error as omitting $\{X^{(b)}, K_J^{(b)}\}$,

$$\Lambda_{(b)} = \max\left\{\frac{w(\tilde{X}, \tilde{K}_J)}{w(X, K_J)}, \frac{w(X, K_J)}{w(\tilde{X}, \tilde{K}_J)}\right\}.$$

Benchmarking is most useful in settings when researchers have strong substantive priors for the types of covariates that are driving the underlying treatment assignment process. We caution that benchmarking cannot be used to rule out the existence of omitted confounders. However, it can be used to help reason about the plausibility of the magnitude of an underlying confounder.

8 Illustration on the magnet example

We apply these methods to the estimates of the Magnet Nursing treatment effect. In the analysis above, we found that Magnet nursing hospitals reduced FTR under the ATT, and reduced both adverse events and FTR under the ATO. Now, the goal is to understand how sensitive these results are to the possible presence of unobserved confounders. We summarize the results for both the marginal sensitivity model and the variance-based sensitivity models in Table 2.

We first analyse the results for the marginal sensitivity model. We report the minimum Λ value at which the effect bounds include zero. As we outlined above, larger values of Λ indicate that it would take larger amounts of unobserved confounding to alter our research conclusions. For the adverse events outcome, the Λ values are 1.06 and 1.02 for the ATT and ATO estimands, respectively. For example, the estimated treatment effect for adverse events under the ATO is −1.7, and when $\Lambda = 1.02$, the bounds on this point estimate are −3.4 and 1.3. These are small departures from a Λ value of 1, the value at which there is no bias from hidden confounders. This indicates

Table 2 Sensitivity values for magnet nursing.

	Adverse events		Failure to rescue	
	MSM	VBM	MSM	VBM
ATT estimand	1.06	0.06	1.8	0.4
ATO estimand	1.04	0.01	1.4	0.3

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that hidden confounding could easily explain this treatment effect. In addition, if we accounted for statistical uncertainty in the bounds, the results would be more conservative. For the FTR outcome, the Λ values are 1.8 and 1.4, for the ATT and ATO estimands respectively. Here, the FTR point estimate is −3.3 points under the ATO, and when $\Lambda = 1.8$, the bounds on this point estimate are −6.1 and 0.20. These values of Λ are much larger than for the adverse event outcome, indicating that the Magnet Hospital treatment effect on the FTR would require considerably larger amounts of hidden bias from unobserved confounders to explain this result compared to the adverse events outcome, though the amount of hidden bias need not be large in absolute terms.

Next, we report results from the variance sensitivity model, similarly estimating the threshold R^2 value at which the effect bounds include zero for both outcomes and estimands. Larger values of R^2 correspond to greater distributional differences between the estimated and oracle weights, implying larger amounts of confounding bias. For the adverse events outcome, the R^2 values are 0.06 and 0.01 for the ATT and ATO estimands, respectively. For the ATO estimand, the bounds on the point estimate are −7.1 and 3.6 at $R^2 = 0.01$. As before, these are quite small values, which indicate that these results could easily be explained by bias from hidden confounders. For the FTR outcome, the R^2 values are 0.4 and 0.3 for the ATT and ATO estimands. The bounds on the point estimate for the ATO are −7.2 and 0.50. These values are much larger than those for the adverse event outcome. Overall, these results are consistent with those from the marginal sensitivity model where the results for adverse events are quite sensitive to bias from hidden confounders and the FTR results are more robust.

Next, we include an amplification and benchmarking analysis for the Magnet Nursing application. We focus this analysis on the failure to rescue outcome for both the ATT and ATO estimands, since it is the largest effect in our analysis. This estimate was also the most robust to bias from a hidden confounder. We also restrict our analysis to the variance-based model, since use of these methods with the marginal sensitivity model are nearly identical in form. We first implemented a benchmarking analysis. For this analysis, we estimated the weights while omitting some subset of the available covariates. Following the formula in Section 7.2, we then estimate the benchmark for each subset of covariates. Given the nature of the imbalances at baseline, we focused on benchmarks for the cluster-level covariates. For the patient-level covariates, we omitted the indicator for whether a patient was transferred as this covariate displayed the largest imbalance at baseline for any patient-level covariate. Finally, we estimated one set of weights where we omitted one cluster-level covariate and one patient-level covariate. For this set of weights, we omitted the indicator for a transfer patient and the percentage of cases that had surgery.

Figure 8 contains the results for both estimands. The demarcated line in the plots represents that values of R^2_U and R^2_V that result from the amplification of R^2 values of 0.4 and 0.3 calculated in Table 2, representing the point where an unobserved confounder would alter our conclusions.

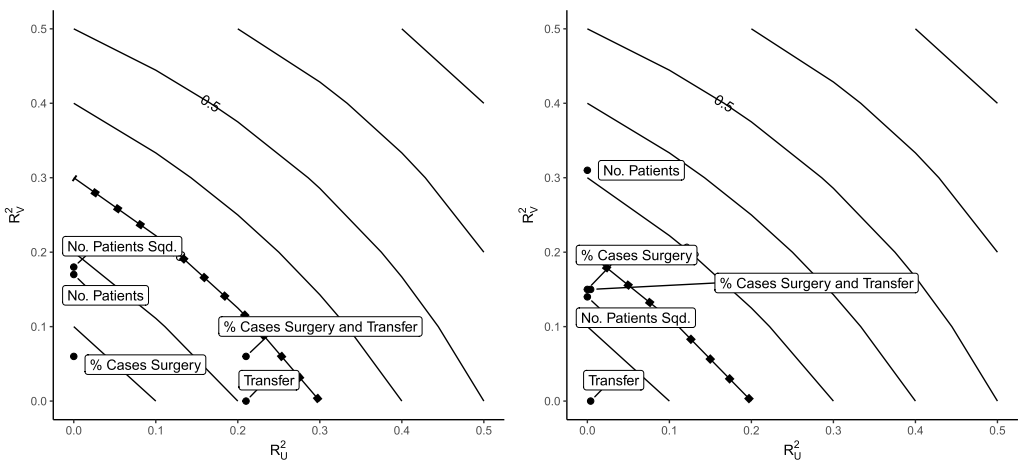

Figure 8 Benchmarking the sensitivity analysis for the variance-based sensitivity model. Dashed-line represents sensitivity analysis upper bound for failure to rescue outcome. The first panel contains results for the ATT estimand. The second panel contains the results for the ATO estimand.

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The benchmark values in the plots represent the values where the observed covariates fall in terms of the R_U^2 and R_V^2 values. For the ATT estimand, none of the benchmark values exceed the R^2 threshold. This implies that a hidden confounder would have to have a larger effect than any of the observed confounders to change our conclusions. For the ATO estimand, only one covariate—patient volume—has a benchmark value that exceeds the R^2 value from the sensitivity analysis. This implies that the unobserved confounder would have to result in an error in the weights that exceeds most of the observed confounders, but less than that of patient volume. Given that patient volume was an extremely strong predictor of Magnet status, this suggests that the unobserved confounder would have to be highly influential to change our substantive conclusions. In general, these results suggest that it would require a very strong unobserved confounder to change the study conclusions for the failure to rescue outcome.

9 Conclusion

In this paper, we developed sensitivity analysis methods for COS designs. Our framework contains several key developments. First, we developed methods for both marginal- and variance-based sensitivity models. Critically, in the COS setting, the marginal sensitivity model is far more conservative than the variance-based methods, but does not yield sharp bounds. In addition, we extended our sensitivity analysis methods to the ATO estimand. The ATO estimand is commonly invoked in cases where overlap is poor and we have found that it is often necessary to deal with overlap violations in COS applications. To our knowledge, this is the first extension of sensitivity analysis methods for weighting estimators to the ATO. We also derived new bias decomposition results that are specific to the COS design. We show that imbalance in cluster-level covariates can amplify any bias in patient-level covariates, which suggests the importance of ensuring that adjustments for cluster-level covariates perform well.

This study also contributes to the literature on the effect of high-quality nursing on patient-level outcomes. Consistent with prior literature, we find that patients have better outcomes at Magnet Nursing hospitals but our study is the first to extend these findings to EGS conditions. Specifically, we found that patients had fewer adverse events and failures to rescue were also lower in Magnet Hospitals. Our study is also the first to use the COS framework to estimate the effect of Magnet Nursing on patient outcomes. We found that despite having a rich data set with a large number of hospitals, covariate overlap was limited, and we could only balance cluster-level covariates well once when we focused on the ATO estimand. Of course, this estimand limits the generalizability of the results. Focusing on the ATO implies that we are estimating the effect of Magnet Nursing certification for the set of hospitals that were on the margin of certification and have overlap between those with and without certification. Finally, we assessed whether these results were sensitive to bias from unobserved confounders. We found that the result for adverse outcomes could easily be explained by hidden confounding, while those for failure to rescue were fairly robust. Overall, the tools we develop here provide analysts with the means to implement sensitivity analyses for the COS design.

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Data availability

The code can be found on our GitHub page at https://github.com/ljk20/cos-sens. We have included detailed instructions in the Readme file on how to reproduce the results. The real data used in our study is proprietary and cannot be shared publicly.

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